

## Year 2: Semester III

### BCA-301 OBJECT ORIENTED PROGRAMMING USING C++

**Introduction to programming paradigms :** Concept of object, class, objects as variables of class data type, difference in structures and class in terms of access to members, private and public members of a class, data & function members. Characteristics of OOP- Data hiding, Encapsulation, data security.

**Basics of C++:** Structure of C++ programs, introduction to defining member functions within and outside a class, keyword using, declaring class, creating objects, constructors & destructor functions, Initializing member values with and without use of constructors, simple programs to access & manipulate data members, cin and cout functions. Dangers of returning reference to a private data member, constant objects and members function, composition of classes, friend functions and classes, using this pointer, creating and destroying objects dynamically using new and delete operators. Static class members, container classes and iterators, proxy classes.

**Operator overloading:** Fundamentals, Restrictions, operator functions as class members v/s as friend functions. Overloading stream function, binary operators and unary operators. Converting between types.

**Inheritance:** Base classes and derived classes, protected members, relationship between base class and derived classes, constructors and destructors in derived classes, public, private and protected inheritance, relationship among objects in an inheritance hierarchy, abstract classes, virtual functions and dynamic binding, virtual destructors.

**Advanced Topics:** Multiple inheritance, virtual base classes, pointers to classes and class members, multiple class members. Templates, exception handling, File handling

**Text Books :** E. Balagursamy : Object oriented programming with C++; TMH Publication  
Deitel and Deitel : C++ How To Program (currently in its 4th edition); PHI

**Reference Books :**

Robert Lafore : Object oriented programming in Turbo C++; Galgotia Publication